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SEMICONDUCTOR PACKAGE

Abstract

A semiconductor package includes a first connection structure having first and second surfaces and including a first redistribution layer, a first semiconductor chip disposed on the first surface and having a first connection pad electrically connected to the first redistribution layer, a second semiconductor chip disposed around the first semiconductor chip on the first surface and having a second connection pad electrically connected to the first redistribution layer, an interconnection bridge disposed on the second surface to be spaced apart from the second surface and connected to the first redistribution layer through a connection member to electrically connect the first and second connection pads to each other, and a second connection structure disposed on the second surface to embed the interconnection bridge and including a second redistribution layer electrically connected to the first redistribution layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application of U.S. patent application Ser. No. 18/528,322, filed Dec. 4, 2023, which is a continuation application of U.S. application Ser. No. 17/863,818, filed Jul. 13, 2022 (now U.S. Pat. No. 11,881,467, issued Jan. 3, 2024), which is a continuation application of U.S. patent application Ser. No. 17/201,538, filed Mar. 15, 2021 (now U.S. Pat. No. 11,410,958, issued Aug. 9, 2022), which is a continuation application of U.S. patent application Ser. No. 16/682,084, filed Nov. 13, 2019, now U.S. Pat. No. 10,985,126, issued Apr. 20, 2021, which claims benefit of priority to Korean Patent Application No. 10-2018-0144383 filed on Nov. 21, 2018 in the Korean Intellectual Property Office, the disclosures of which are incorporated herein in their entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a semiconductor package.

BACKGROUND

[0003] The interposer market is growing, due to high-specifications of a set and the adoption of high bandwidth memory (HBM). Currently, silicon is a main material of an interposer. For example, a semiconductor package using an interposer is manufactured by surface-mounting a die on a silicon-based interposer and molding the surface-mounted die with a molding material.

SUMMARY

[0004] An aspect of the present disclosure is to provide a semiconductor package which includes a connection structure in which an interconnection bridge, capable of replacing a conventional interposer, is embedded. In spite of the embedded interconnection bridge, the semiconductor package may control warpage and may be excellent in yield and processability, low in cost, and free of process and yield issues on a fine pattern.

[0005] One proposal of the present disclosure is to provide a semiconductor package in which a

first connection structure is directly disposed on a plurality of semiconductor chips, an interconnection bridge is surface-mounted on the first connection structure to electrically connect the plurality of semiconductor chips to each other, and a second connection structure is disposed on the first connection structure to embed the interconnection bridge.

[0006] According to an aspect of the present disclosure, a semiconductor package includes a first connection structure having a first surface and a second surface opposing the first surface and including a first redistribution layer, a first semiconductor chip, disposed on the first surface of the first connection structure, having a first connection pad electrically connected to the first redistribution layer, a second semiconductor chip, disposed around the first semiconductor chip on the first surface of the first connection structure, having a second connection pad electrically connected to the first redistribution layer, an interconnection bridge disposed on the second surface of the first connection structure to be spaced apart from the second surface of the first connection structure and connected to the first redistribution layer through a connection member to electrically connect the first and second connection pads to each other, and a second connection structure, disposed on the second surface of the first connection structure to embed the interconnection bridge, including a second redistribution layer electrically connected to the first redistribution layer.

[0007] According to an aspect of the present disclosure, a semiconductor package includes a first semiconductor chip having a first active surface on which a first connection pad is disposed, a second semiconductor chip, disposed around the first semiconductor chip, having a second active surface on which a second connection pad is disposed, a first connection structure including a first insulating layer disposed on the first and second active surfaces, a first redistribution layer disposed on the first insulating layer, and first connection vias penetrating through the first insulating layer and respectively, electrically connecting the first redistribution layer to the first and second connection pads, an interconnection bridge disposed on the first connection structure and connected to the first redistribution layer to electrically connect the first and second connection pads to each other, and a second connection structure including a second insulating layer disposed on the first connection structure to cover at least a portion of the first redistribution layer and to embed the interconnection bridge, a second redistribution layer disposed on the second insulating layer, and a second connection structure penetrating through the second insulating layer and electrically connecting the first and second redistribution layers to each other. The first connection vias are in physical contact with the respective first and second connection pads.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0009] FIG. 1 is a block diagram schematically illustrating an example of an electronic device system;

[0010] FIG. 2 is a schematic perspective view illustrating an example of an electronic device;

[0011] FIG. 3 is a schematic cross-sectional view illustrating a case in which a 3D BGA package is mounted on a mainboard of an electronic device;

[0012] FIG. 4 is a schematic cross-sectional view illustrating a case in which a 2.5D silicon interposer package is mounted on a mainboard;

[0013] FIG. 5 is a schematic cross-sectional view illustrating a case in which a 2.5D organic interposer package is mounted on a mainboard;

[0014] FIG. 6 is a schematic cross-sectional view illustrating an example of a semiconductor package;

[0015] FIG. 7 is a cutaway plan view taken along line I-I' of the semiconductor package in FIG. 6; [0016] FIGS. 8 and 9 are schematic process diagrams illustrating an example of manufacturing the semiconductor package in FIG. 6; [0017] FIG. 10 is a schematic cross-sectional view illustrating another example of a semiconductor package; and [0018] FIG. 11 is a cutaway plan view taken along line II-II' of the semiconductor package in FIG. 10.

DETAILED DESCRIPTION

[0019] Hereinafter, embodiments of the present disclosure will be described as follows with reference to the attached drawings.

Electronic Device

[0020] FIG. 1 is a schematic block diagram illustrating an example of an electronic device system. [0021] Referring to FIG. 1, an electronic device **1000** may accommodate a mainboard **1010** therein. The mainboard **1010** may include chip related components **1020**, network related components **1030**, other components **1040**, and the like, physically or electrically connected thereto. These components may be connected to others to be described below to form various signal lines **1090**. [0022] The chip related components **1020** may include a memory chip such as a volatile memory (for example, a dynamic random access memory (DRAM)), a non-volatile memory (for example, a read only memory (ROM)), a flash memory, or the like; an application processor chip such as a central processor (for example, a central processing unit (CPU)), a graphics processor (for example, a graphics processing unit (GPU)), a digital signal processor, a cryptographic processor, a microprocessor, a microcontroller, or the like; and a logic chip such as an analog-to-digital (ADC) converter, an application-specific integrated circuit (ASIC), or the like. However, the chip related components **1020** are not limited thereto, but may also include other types of chip related components. In addition, the chip related components **1020** may be combined with each other. [0023] The network related components **1030** may include protocols such as wireless fidelity (Wi-Fi) (Institute of Electrical And Electronics Engineers (IEEE) 802.11 family, or the like), worldwide interoperability for microwave access (WiMAX) (IEEE 802.16 family, or the like), IEEE 802.20, long term evolution (LTE), evolution data only (Ev-DO), high speed packet access+(HSPA+), high speed downlink packet access+(HSDPA+), high speed uplink packet access+(HSUPA+), enhanced data GSM environment (EDGE), global system for mobile communications (GSM), global positioning system (GPS), general packet radio service (GPRS), code division multiple access (CDMA), time division multiple access (TDMA), digital enhanced cordless telecommunications (DECT), Bluetooth, 3G, 4G, and 5G protocols, and any other wireless and wired protocols, designated after the abovementioned protocols. However, the network related components **1030** are not limited thereto, but may also include a variety of other wireless or wired standards or protocols. In addition, the network related components **1030** may be combined with each other, together with the chip related components **1020** described above. [0024] Other components **1040** may include a high frequency inductor, a ferrite inductor, a power inductor, ferrite beads, a low temperature co-fired ceramic (LTCC), an electromagnetic interference (EMI) filter, a multilayer ceramic capacitor (MLCC), or the like. However, other components **1040** are not limited thereto, but may also include passive components used for various other purposes, or the like. In addition, other components **1040** may be combined with each other, together with the chip related components **1020** or the network related components **1030** described above. [0025] Depending on a type of the electronic device **1000**, the electronic device **1000** may include other components that may or may not be physically or electrically connected to the mainboard **1010**. These other components may include, for example, a camera **1050**, an antenna **1060**, a display **1070**, a battery **1080**, an audio codec (not illustrated), a video codec (not illustrated), a power amplifier (not illustrated), a compass (not illustrated), an accelerometer (not illustrated), a gyroscope (not illustrated), a speaker (not illustrated), a mass storage unit (for example, a hard disk

drive) (not illustrated), a compact disk (CD) drive (not illustrated), a digital versatile disk (DVD) drive (not illustrated), or the like. However, these other components are not limited thereto, but may also include other components used for various purposes depending on a type of electronic device **1000**, or the like.

[0026] The electronic device **1000** may be a smartphone, a personal digital assistant (PDA), a digital video camera, a digital still camera, a network system, a computer, a monitor, a tablet PC, a laptop PC, a netbook PC, a television, a video game machine, a smartwatch, an automotive component, or the like. However, the electronic device **1000** is not limited thereto, but may be any other electronic device processing data.

[0027] FIG. **2** is a schematic perspective view illustrating an example of an electronic device.

[0028] Referring to FIG. **2**, a semiconductor package may be used for various purposes in the various electronic devices **1000** as described above. For example, a motherboard **1110** may be accommodated in a body **1101** of a smartphone **1100**, and various electronic components **1120** may be physically or electrically connected to the motherboard **1110**. In addition, other components that may or may not be physically or electrically connected to the motherboard **1110**, such as a camera module **1130**, may be accommodated in the body **1101**. Some of the electronic components **1120** may be the chip related components, for example, a semiconductor package **1121**, but are not limited thereto. The electronic device is not necessarily limited to the smartphone **1100**, but may be other electronic devices as described above.

Semiconductor Package

[0029] Generally, numerous fine electrical circuits are integrated in a semiconductor chip.

However, the semiconductor chip may not serve as a finished semiconductor product in itself, and may be damaged due to external physical or chemical impacts. Therefore, the semiconductor chip itself may not be used, but may be packaged and used in an electronic device, or the like, in a packaged state.

[0030] Here, semiconductor packaging is required due to the existence of a difference in a circuit width between the semiconductor chip and a mainboard of the electronic device in terms of electrical connections. In detail, a size of connection pads of the semiconductor chip and an interval between the connection pads of the semiconductor chip are very fine, but a size of component mounting pads of the mainboard used in the electronic device and an interval between the component mounting pads of the mainboard are significantly larger than those of the semiconductor chip. Therefore, it may be difficult to directly mount the semiconductor chip on the mainboard, and packaging technology for buffering a difference in a circuit width between the semiconductor chip and the mainboard is required.

[0031] Hereinafter, among semiconductor packages manufactured by such a packaging technology, a semiconductor package using an interposer will be described more fully with reference to accompanying drawings.

[0032] FIG. **3** is a schematic cross-sectional view illustrating a case in which a 3D BGA package is mounted on a mainboard of an electronic device.

[0033] Among semiconductor chips, an application specific integrated circuit (ASIC) such as a graphics processing unit (GPU) is very expensive. Therefore, it is significantly important to package ASICs at high yield. In this regard, a ball grid array (BGA) may be prepared to redistribute thousands to hundreds of thousands of connection pads before mounting a semiconductor chip. An expensive semiconductor chip, such as a GPU **2220** or the like, may be mounted on a BGA substrate **220** using a surface-mount technology (SMT) or the like, and then is packaged. Ultimately, the packaged semiconductor chip may be mounted on a mainboard **2110**.

[0034] In the case of a GPU **2220**, it is necessary to significantly reduce a signal path with a memory such a high bandwidth memory (HBM). To this end, a semiconductor chip such as the HBM **2220** may be packaged after being mounted on an interposer **2230**. The packaged semiconductor chip may be laminated on a package, on which the GPU **2220** is mounted, in a

package-on-package (POP) configuration. Unfortunately, a thickness of a device may be significantly increased, and there may be a limitation in significantly reducing the signal path. [0035] FIG. 4 is a schematic cross-sectional view illustrating a case in which a 2.5D silicon interposer package is mounted on a mainboard.

[0036] In order to address the above issues, a semiconductor package **2310**, including an organic interposer, may be manufactured using a 2.5D interposer technology in which a first semiconductor chip such as a GPU **2220** and a second semiconductor chip such as a HBM **2240** are packaged after being surface-mounted side by side on a silicon interposer **2250**. In this case, the GPU **2220** and the HBM **2240**, having thousands to hundreds of thousands of connection pads, may be redistributed through the interposer **2250** and may be electrically connected via a shortest path. When the semiconductor package **2310**, including such an interposer, is redistributed after being mounted again on the BGA substrate **2210**, or the like, the semiconductor package **2310** may be ultimately mounted on the mainboard **2110**. Since it is significantly difficult to form a through-silicon via (TSV) and manufacturing cost of the silicon interposer **2250** is significantly high, the silicon interposer **2250** may be disadvantageous in larger area and low cost.

[0037] FIG. 5 is a schematic cross-sectional view illustrating a case in which a 2.5D organic interposer package is mounted on a mainboard.

[0038] In order to address the above issues, an organic interposer **2260** may be used rather than the silicon interposer **2250**. For example, a semiconductor package **2320**, including an organic interposer, may be manufactured using a 2.5D interposer technology in which a first semiconductor chip such as a GPU **2220** and a second semiconductor chip such as an HBM **2240** are packaged after being surface-mounted side by side on the organic interposer **2260**. In this case, the GPU **2220** and the HBM **2240**, having thousands to hundreds of thousands of connection pads, may be redistributed through the interposer **2260** and may be electrically connected via a shortest path. When the semiconductor package **2310**, including such an organic interposer, is redistributed after being mounted on a BGA substrate **2210**, or the like, the semiconductor package **2310** may be ultimately mounted on the mainboard **2110**. In addition, the semiconductor package **2310**, including such an organic interposer, may be advantageous in large area and low cost. However, the semiconductor package **2310**, including such an organic interposer, may suffer from occurrence of warpage, poor underfill resin filling, occurrence of cracking between a die and a molding material, and the like, due to a mismatch of coefficients of thermal expansion (CTE) of the interposer **2260** and chips **2220** and **2240** when a molding process is performed. Moreover, such an organic interposer may be disadvantageous in implementing a fine pattern.

[0039] In order to address the above issues, although not illustrated in detail, an interconnection bridge having a fine pattern may be additionally formed and inserted into a cavity of a BGA substrate to be embedded. However, since it is difficult to form the cavity and to implement a corresponding fine circuit in the BGA substrate, process and yield issues may be reduced. Accordingly, there is a need for a novel semiconductor package capable of addressing all the issues.

[0040] FIG. 6 is a schematic cross-sectional view illustrating an example of a semiconductor package, and FIG. 7 is a cutaway plan view taken along line I-I' of the semiconductor package in FIG. 6.

[0041] Referring to FIGS. 6 and 7, a semiconductor package **100A** according to an example embodiment includes a first connection structure **110** having a first surface and a second surface opposing the first surface and including a first redistribution layer **112**, a first semiconductor chip **121**, disposed on the first surface of the first connection structure **110**, having a first connection pad **121P** electrically connected to the first redistribution layer **112**, a second semiconductor chip **122**, disposed around the first semiconductor chip **121** on the first surface of the first connection structure **110**, having a second connection pad **122P** electrically connected to the first redistribution layer **112**, an interconnection bridge **150** disposed on the second surface of the first connection

structure **110** to be spaced apart from the second surface of the first connection structure **110** and connected to the first redistribution layer **112** through a connection member **153** to electrically connect the first and second connection pads **121P** and **122P** to each other, and a second connection structure **140**, disposed on the second surface of the first connection structure **110** to embed the interconnection bridge **150**, including a second redistribution layer **142** electrically connected to the first redistribution layer **112**.

[0042] In addition, the semiconductor package **100A** may further include an encapsulant **130** disposed on the first surface of the first connection structure **110** to cover at least a portion of each of the first and second semiconductor chips **121** and **122**, an insulating material **155** disposed between the second surface of the first connection structure **110** and the interconnection structure **150** to embed at least a portion of the connection member **153**, a first electrical connection metal **160** disposed on a side, opposing a side on which the first connection structure **110** is disposed, and electrically connected to the second redistribution layer **142**.

[0043] As described above, in the semiconductor package **100A**, the interconnection bridge **150**, having a fine pattern, is manufactured in advance and then is located on the first redistribution layer **112** of the first connection structure **110** in a surface-mount format using a low melting point metal, for example, a connection member **153**, including tin (Sn) or a Sn-containing alloy, to be electrically connected. According to the semiconductor package **100A**, process and yield issues on a fine pattern substantially do not occur. In addition, due to use of the interconnection bridge **150** having a fine pattern, the number of layers of the respective first and second redistribution layers **112** and **142** of the first and second connection structures **110** and **140** may be significantly decreased, and each of the first and second redistribution layers **112** and **142** may be provided as, for example, only one layer. Thus, improved yield and processability may be achieved.

[0044] The semiconductor package **100A** may have a structure in which the first connection structure **110** is directly formed on the first and second semiconductor chips **121** and **122**. For example, the first connection structure **110** may include a first insulating layer **111** disposed on the first and second semiconductor chips **121** and **122**, a first redistribution layer **112** disposed on the first insulating layer **111**, and a first connection via **113** penetrating through the first insulating layer **111** and electrically connecting the first redistribution layer **112** to the first and second connection pads **121P** and **122P**. The first connection via **113** may be in physical contact with the respective first and second connection vias **121P** and **122P**. For example, the first connection via **113** may directly connect the first redistribution layer **112** to the first and second connection pads **121P** and **122P**. In this case, the first connection structure **110** may be formed using the first and second semiconductor chips **121** and **122**, each having a significant thickness, as a base, which is effective in controlling process warpage. In addition, since the first connection via **113** may be directly formed on the first and second connection pads **121P** and **122P** without formation of an additional bump, manufacturing cost may be reduced and process difficulty may be lowered, which is advantageous in processability.

[0045] On the other hand, a semiconductor package **100A** according to an example embodiment may have a structure in which a second connection structure **140** is directly disposed on the first connection structure **110**. For example, the second connection structure **140** may include a second insulating layer **141**, disposed on the second surface of the first connection structure **110**, covering at least a portion of the interconnection bridge **150**, a second redistribution layer **142** disposed on the second insulating layer **141**, and a second connection via **143** penetrating through the second insulating layer **141** and electrically connecting the first and second redistribution layers **112** and **142** to each other. The second connection via **143** may be in physical contact with the respective first and second redistribution layers **112** and **142**. For example, the second connection via **143** may directly connect the first and second redistribution layers **112** and **142** to each other, which may be more effective in controlling process warpage. In addition, the interconnection bridge **150** may be embedded in the second connection structure **140** to be effectively protected and may be thinned.

For example, the interconnection bridge **150** may be disposed on a level between surfaces, respectively being in contact with the first and second redistribution layers **112** and **142** of the second connection via **143**.

[0046] Hereinafter, each component included in the semiconductor package **100A** according to an example embodiment will be described more fully with reference to accompanying drawings.

[0047] The first connection structure **110** may primarily redistribute the first and second pads **121P** and **122P** of the first and second semiconductor chips **121** and **122**. Tens to hundreds of thousands of connection pads, having various functions, may be redistributed through the first connection structure **110**, and may be physically and/or electrically connected to other components depending on the functions thereof. The first connection structure **110** may include a first insulating layer disposed on first and second active surfaces of the first and second semiconductor chips **121** and **122**, a first redistribution layer **112** disposed on the first insulating layer **111**, and a first connection via **113**, penetrating through the first insulating layer **111**, connected to the first redistribution layer **112**. The first connection via **113** may also be connected to the first and second connection pads **121P** and **122P**. The first insulating layer **111**, the first redistribution layer **112**, and the first connection via **113**, constituting the first connection structure **110**, may be greater in number than those illustrated in the drawings, but may be provided as, for example, only one layer as illustrated in the drawings to secure processability and yield and to implement thinning.

[0048] A material of the first insulating layer **111** may be an insulating material. The insulating material may be a photosensitive material such as a photoimageable dielectric (PID). For example, the first insulating layer **111** may be a photosensitive layer. When the first insulating layer **111** has photosensitive properties, the first insulating layer **111** may be further thinned and a fine pitch of the first connection via **113** may be more easily achieved.

[0049] The first redistribution layer **112** may serve to primarily redistribute the first and second connection pads **121P** and **122P**, and may be formed of a metal material such as copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or alloys thereof. The first redistribution layer **112** may perform various functions depending on a design of a corresponding layer. For example, the first redistribution layer **112** may include a ground pattern, a power pattern, a signal pattern, and the like. The signal pattern may include various signals, such as data signal and the like, except for the ground pattern, the power pattern, and the like. In addition, the first redistribution layer **112** may include a connection via pad having various uses.

[0050] The first connection via **113** may electrically connect the first redistribution layer **112** to the first and second connection pads **121P** and **122P**. As a result, an electrical path is formed. A material of the first connection via **113** may be a metal material such as copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or alloys thereof. The first connection via **143** may be a filled-type via filled with the metal material, or a conformal-type via in which a metal material is formed along a wall surface of a via hole. Moreover, the first connection via **143** may have a tapered shape or the like. The first connection via **113** may also include a via for signal, a via for grounding, a via for power, and the like. The via for grounding and the via for power may be the same via. The first connection via **113** may be in physical contact with the first and second connection pads **121P** and **122P**.

[0051] Each of the first and second semiconductor chips **121** and **122** may also be an integrated circuit (IC) in which hundreds to millions of or more elements are integrated in a single chip. A base material of a body of each of the first and second semiconductor chips **121** and **122** may also be silicon (Si), germanium (Ge), gallium arsenide (GaAs), or the like. Various circuits may be formed on the body of each of the first and second semiconductor chips **121** and **122**. The first and second connection pads **121P** and **122P** may electrically connect the first and second semiconductor chips **121** and **122** to other components.

[0052] A material forming the first and second connection pads **121P** and **122P** may also be a metal material such as copper (Cu), aluminum (Al), or the like, but is not limited thereto. A passivation

layer, not illustrated, may be formed on the body of each of the first and second semiconductor chips **121** and **122** to expose the first and second connection pad **121P** and **122P**. The passivation layer may be an oxide layer, a nitride layer, or the like, or a double layer of an oxide layer and a nitride layer. An insulating layer, not illustrated, or the like may be further disposed at a necessary position. The first and second semiconductor chips **121** and **122** may have first and second active surfaces, on which the first and second connection pads **121P** and **122P** are disposed, and first and second inactive surfaces opposing the first and second active surfaces, respectively.

[0053] The first semiconductor chip **121** may be an application specific integrated circuit (ASIC) such as a graphics processing unit (GPU). The second semiconductor chip **122** may be a stacked memory such as a high bandwidth memory (HBM). For example, the first and second semiconductor chips **121** and **122** may be expensive chips, each having hundreds of thousands of I/O, but are not limited thereto. A greater number of second semiconductor chips **122** than first semiconductor chips **121** may be disposed, and may be disposed around semiconductor chips **121**, the first respectively. For example, two second semiconductor chips may be disposed on both sides of the first semiconductor chip **121**, respectively. However, this is merely an example and is not limited thereto.

[0054] The encapsulant **130** may protect the first and second semiconductor chips **121** and **122**. An encapsulation form is not limited as long as the encapsulant **130** covers at least a portion of each of the first and second semiconductor chips **121** and **122**. For example, the encapsulant **130** may cover first and second inactive surfaces of the first and second semiconductor chips **121** and **122** and at least a portion of a side surface of each of the first and second semiconductor chips **121** and **122**. The encapsulant **130** may include an insulating material. The insulating material may be a material including an inorganic filler and an insulating resin, for example, a thermosetting resin such as an epoxy resin, a thermoplastic resin such as a polyimide resin, a resin having a reinforcing material, such as an inorganic filler impregnated in the thermosetting resin and the thermoplastic resin, such as ABF, FR-4, BT, or the like. Alternatively, a photoimageable encapsulant (PIE) may be used as the insulating material. As necessary, a material, in which a thermosetting resin or a thermoplastic resin is impregnated with an inorganic filler and/or a core material such as a glass fiber, for example, prepreg may also be used as the insulating material.

[0055] The second connection structure **140** may provide an electrical connection path for electrically connecting the first and second connection pads **121P** and **122P** of the first and second semiconductor chips **121** and **122** to an external component such as a printed circuit board (PCB) **200** or the like. The second connection structure **140** may serve to protect an interconnection bridge **150** by embedding the interconnection bridge. The second connection structure **140** includes a second insulating layer **141** disposed on the first connection structure **110**, a second redistribution layer **142** disposed on the second insulating layer **141**, and a second connection via **143**, penetrating through the second insulating layer **141**, connected to the second redistribution layer **142**. The second connection via **143** may also be connected to the first redistribution layer **112**. The second insulating layer **141**, the second redistribution layer **142**, and the second connection via **143**, constituting the first connection structure **110**, may be greater in number than those illustrated in the drawings, but may be provided as, in detail, only one layer as illustrated in the drawings to secure processability and yield and to implement thinning.

[0056] A material of the second insulating layer **141** may be an insulating material. The insulating material may be a non-photosensitive material such as ABF including an insulating resin and an inorganic filler. For example, the second insulating layer **141** may be a non-photosensitive insulating layer. In this case, the second insulating layer **141** may cover the interconnection bridge **150** in the form of a film having a significantly great thickness to stably embed the interconnection bridge **150**. In addition, process warpage may be effectively controlled.

[0057] The second redistribution layer **142** may provide a pad region of the first electrical connection metal **160** for electrically connecting the first and second connection pads **121P** and

122P to an external component. The second redistribution layer **142** may be formed of a metal material such as copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or alloys thereof. The second redistribution layer **142** may perform various functions depending on a design of a corresponding layer. For example, the second redistribution layer **142** may include a ground pattern, a power pattern, a signal pattern, and the like. The signal pattern may include various signals, such as data signal and the like, except for the ground pattern, the power pattern, and the like. In addition, the second redistribution layer **142** may include a via pad and/or an electrical connection metal pad having various uses.

[0058] The second connection via **143** may electrically connect the first and second redistribution layers **112** and **142**. As a result, a vertical electrical path is formed. A material of the second connection via **143** may be a metal material such as copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or alloys thereof. The second connection via **143** may be a filled-type via filled with the metal material, or a conformal-type via in which a metal material is formed along a wall surface of a via hole. Moreover, the second connection via **143** may have a tapered shape or the like. The second connection via **143** may also include a via for signal, a via for grounding, a via for power, and the like. The via for grounding and the via for power may be the same via. The second connection via **143** may be in physical contact with the first and second connection pads **121P** and **122P**.

[0059] The interconnection bridge **150** electrically connects the first and second connection pads **121P** and **122P**. For example, the interconnection bridge **150** may electrically connect signal pads of the first and second connection pads **121P** and **122P** to each other. The interconnection bridge **150** may include a fine pattern formed therein. For example, the interconnection bridge **150** may include a body, having an active surface, on which a connection electrode **150P** is disposed, and an inactive surface opposing the active surface, and a connection bump **150B** disposed on the connection electrode **150P** to be electrically connected to the connection member **153**. The body may be a silicon-based body in which a fine pattern for interconnection is formed. For example, the interconnection bridge **150** may be a silicon-based interconnection bridge in which a fine circuit is designed. The number of interconnection bridges **150** is not limited and may be greater or smaller than the number of those illustrated in the drawings.

[0060] The interconnection bridge **150** is disposed on the first redistribution layer **112** of the first connection structure **110** through the connection member **153** in a surface-mount form. For example, the first redistribution layer **112** of the first connection structure **110** and the interconnection bridge **150** may be spaced apart from each other, and a space therebetween may be filled with an insulating material **155**. The insulating material **155** may embed the connection member **153** and may fix the interconnection bridge **150**. The insulating material **155** has a region, covering a bottom surface of the first connection structure **110**, and a region covering a top surface of the interconnection bridge **150**. The region, covering a bottom surface of the first connection structure **110**, may have an area larger than an area of the region covering a top surface of the interconnection bridge **150**. For example, the insulating material **155** may be applied on a bottom surface of the first connection structure **110** to fix the interconnection bridge **150**, and may have a tapered shape, in which a top surface has a width greater than a width of a bottom surface, when viewed from above. Each of the connection electrode **150P** and the connection bump **150B** may include a known metal material such as copper (Cu), aluminum (Al), or the like. The connection member **150** may include a low melting point metal, for example, tin (Sn) or a Sn-containing alloy. For example, the connection member **150** may be a solder. The insulating material **155** may be an underfill resin including epoxy or the like.

[0061] The first electrical connection metal **160** may physically and/or electrically connect the semiconductor package **100A** to an external component. For example, the semiconductor package **100A** may be mounted on a BGA substrate, or the like, through the first electrical connection metal **160**. The first electrical connection metal **160** may be formed of a low melting point metal, for

example, a solder or the like, but is merely an example and a material thereof is not limited thereto. The first electrical connection metal **160** may be a land, a ball, a pin, or the like. The first electrical connection metal **160** may be formed of multiple layers or a single layer. The first electrical connection metal **160** may include a copper pillar and a solder when the first electrical connection metal **160** is formed of multiple layers, and may include a tin-silver solder or copper when the first electrical connection metal **160** is formed of a single layer, but is not limited thereto. The number, an interval, a disposition form or the like, of the first electrical connection metal **160** is not limited, and may be sufficiently modified by a person skilled in the art, depending on design. For example, the number of first electrical connection metals **160** may be several tens to several thousands, depending on the number of the first and second connection pads **121P** and **122P**. However, the number of first electrical connection metals **160** may be several thousands or more or several thousands or less.

[0062] The printed circuit board (PCB) **200** may be disposed on a side opposing a side on which the first connection structure **110** of the second connection structure **140** of the semiconductor package **100A** is disposed, and may be connected to the first electrical connection metal **160**. For example, the semiconductor package **100A** may be surface-mounted on the PCB **200** through the first electrical connection metal **160**. The PCB **200** may be mounted on a mainboard of an electrical device, or the like, via the second electrical connection metal **210**. The PCB **200** may be a BGA substrate, but is not limited thereto. As necessary, the semiconductor package **100A** may be manufactured while the PCB **200** and/or the second electrical connection metal **210** are disposed thereon. For example, the semiconductor package **100A** may include the PCB **200** and/or the second electrical connection metal **210**.

[0063] FIGS. **8** and **9** are schematic process diagrams illustrating an example of manufacturing the semiconductor package in FIG. **6**.

[0064] Referring to FIG. **8**, first and second semiconductor chips **121** and **122** are disposed on a tape, not illustrated, or the like in a face-down orientation. The first and second semiconductor chips **121** and **122** are encapsulated by an encapsulant **130** using a manner such as ABF lamination and curing, or the like. By using the first and second semiconductor chips **121** and **122**, encapsulated by the encapsulant **130**, as a base, a first insulating layer **111** is formed on first and second active surfaces of the respective first and second semiconductor chips **121** and **122** by PID coating and curing. After a via hole is formed by a photolithography process, a first redistribution layer **112** and a first connection via **113** are formed by a plating process, for example, an additive process (AP), a semi-AP (SAP), a modified SAP (MSAP), tenting, or the like. For example, a first connection structure **110** is directly formed on the first and second semiconductor chips **121** and **122**. In this case, the first connection via **113** may be in physical contact with the first and second connection pads **121P** and **122P**. An interconnection bridge **150**, capable of electrically connecting the first and second connection pads **121P** and **122P**, is mounted on the first redistribution layer **112** via a connection member **153**, and is fixed using an insulating material **155**.

[0065] Referring to FIG. **9**, a second insulating layer **141** is formed on the first connection structure **110** to embed the interconnection bridge **150** using a manner such as ABF lamination and curing, or the like. After a via hole is formed by laser drilling or the like, a second redistribution layer **142** and a second connection via **143** are formed by the above-mentioned plating process. For example, the second connection structure **140** is directly formed on the first connection structure **110** to embed the interconnection bridge **150**. A first electrical connection metal **160** is formed to be connected to the second redistribution layer **142**. Through a series of processes, the above-described semiconductor package **100A** is manufactured. As necessary, the manufactured semiconductor package **100A** may be surface-mounted and disposed on a printed circuit board (PCB) **200**, having a second electrical connection metal **210**, via the first electrical connection metal **160**.

[0066] FIG. **10** is a schematic cross-sectional view illustrating another example of a semiconductor package, and FIG. **11** is a cutaway plan view taken along line II-II' of the semiconductor package in

FIG. 10.

[0067] Referring to FIGS. **10** and **11**, a semiconductor package **100B** according to another example embodiment further includes one or more passive components **180**, disposed on a first redistribution layer **112** of a first connection structure **110**, as compared to the above-described semiconductor package **100A**. Each of the passive components **180** is embedded in a second insulating layer **141** of a second connection structure **140**. Each of the passive components **180** is electrically connected to at least one of first and second connection pads **121P** and **122P** through the first redistribution layer **112**. Each of the passive components **180** may include a capacitor and/or an inductor. The other descriptions are substantially the same as the detailed description of the package connection system **100B** according to another example embodiment, and will be omitted herein.

[0068] As described above, example embodiments provide a semiconductor package which includes a connection structure in which an interconnection bridge, capable of replacing a conventional interposer, is embedded. In spite of the embedded interconnection bridge, the semiconductor package may control warpage and may be excellent in yield and processability, low in cost, and free of process and yield issues on a fine pattern.

[0069] In the present disclosure, the terms “lower side”, “lower portion”, “lower surface,” and the like, have been used to indicate a direction toward a mounted surface of the electronic component package in relation to cross sections of the drawings, the terms “upper side”, “upper portion”, “upper surface,” and the like, have been used to indicate an opposite direction to the direction indicated by the terms “lower side”, “lower portion”, “lower surface,” and the like. However, these directions are defined for convenience of explanation only, and the claims are not particularly limited by the directions defined, as described above.

[0070] The meaning of a “connection” of a component to another component in the description includes an indirect connection through an adhesive layer as well as a direct connection between. In addition, “electrically connected” means two components, including a physical connection and a physical disconnection. It can be understood that when an element is referred to as “first” and “second”, the element is not limited thereby. These terms may be used only for a purpose of distinguishing the element from the other elements, and may not limit the sequence or importance of the elements. In some cases, a first element may be referred to as a second element without departing from the scope of the claims set forth herein. Similarly, a second element may also be referred to as a first element.

[0071] The term “an example embodiment” used herein does not always refer to the same example embodiment, and is provided to emphasize a particular feature or characteristic different from that of another example embodiment. However, example embodiments provided herein are considered to be able to be implemented by being combined in whole or in part one with another. For example, one element described in a particular example embodiment, even if it is not described in another example embodiment, may be understood as a description related to another example embodiment, unless an opposite or contradictory description is provided therein.

[0072] Terms used herein are used only in order to describe an example embodiment rather than to limit the present disclosure. In this case, singular forms include plural forms unless necessarily interpreted otherwise, based on a particular context.

[0073] While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A semiconductor package comprising: a first connection structure comprising a first insulating layer, a first redistribution layer below the first insulating layer, and a first connection via

penetrating from the first redistribution layer into the first insulating layer; a first semiconductor chip on the first connection structure and a second semiconductor chip each on the first connection structure and each comprising connection pads electrically connected to a corresponding one of the first redistribution layer through the first connection via; an interconnection bridge below the first connection structure, the interconnection bridge electrically connecting the first and the second semiconductor chips through the first redistribution layer; an insulating material between the interconnection bridge and the first connection structure; and a second connection structure below the first connection structure, the second connection structure comprising a second insulating layer covering the first redistribution layer and the insulating material, a second redistribution layer below the second insulating layer, and a second connection via penetrating through the second insulating layer and electrically connecting the second redistribution layer to a corresponding one of the first redistribution layer, wherein the first connection via is tapered in a direction toward the connection pads of the first and the second semiconductor chips, wherein the first redistribution layer is disposed on a lower surface of the first insulating layer, and wherein the second insulating layer contacts a side surface of the first redistribution layer and a lower surface of the first redistribution layer.

2. The semiconductor package of claim 1, wherein the first redistribution layer includes a first group electrically connecting the first semiconductor chip to the interconnection bridge, a second group electrically connecting the second semiconductor chip to the interconnection bridge, and a third group electrically connecting the first and the second semiconductor chips to the second redistribution layer.

3. The semiconductor package of claim 2, wherein the first connection via electrically connects the first and the second groups of the first redistribution layer to the first and the second semiconductor chips.

4. The semiconductor package of claim 2, wherein the second connection via electrically connects the second redistribution layer to the third group of the first redistribution layer.

5. The semiconductor package of claim 2, wherein the second insulating layer contacts a side surface of the third group of the first redistribution layer and a lower surface of the third group of the first redistribution layer.

6. The semiconductor package of claim 1, wherein the interconnection bridge overlaps the first semiconductor chip and the second semiconductor chip in a vertical direction.

7. The semiconductor package of claim 1, wherein the interconnection bridge comprises a body having a first surface on which a connection electrode is disposed, and a second surface opposite to the first surface, and wherein the first surface of the interconnection bridge faces the first connection structure.

8. The semiconductor package of claim 7, further comprising: a connection member electrically connecting the connection electrode of the interconnection bridge to the first redistribution layer.

9. The semiconductor package of claim 8, wherein the insulating material surrounds at least a portion of the connection member.

10. The semiconductor package of claim 8, wherein the interconnection bridge further comprises a connection bump disposed on the connection electrode, and wherein the connection bump contacts the connection member.

11. The semiconductor package of claim 1, wherein the second connection via overlaps one of the first semiconductor chip and the second semiconductor chip.

12. The semiconductor package of claim 1, wherein an area of the interconnection bridge is greater than an area between the first semiconductor chip and the second semiconductor chip.

13. The semiconductor package of claim 1, wherein a height of the second connection via is greater than a height of the first connection via.

14. The semiconductor package of claim 1, wherein a width of the second connection via is greater than a width of the first connection via.

15. A semiconductor package comprising: a first connection structure comprising a first insulating layer, a first redistribution layer below the first insulating layer, and a first connection via extending from the first redistribution layer into the first insulating layer; a plurality of semiconductor chips on the first connection structure, the plurality of semiconductor chips comprising connection pads electrically connected to the first redistribution layer; an interconnection bridge below the first connection structure, the interconnection bridge electrically connecting the plurality of semiconductor chips through the first redistribution layer; an underfill material between the interconnection bridge and the first connection structure; and a second connection structure comprising a second redistribution layer below the interconnection bridge, and a second connection via connecting the second redistribution layer to the first redistribution layer, wherein the first connection via is tapered in a direction toward the connection pads of the plurality of semiconductor chips.

16. A semiconductor package comprising: a first connection structure comprising a first redistribution layer, and a first connection via electrically connected to the first redistribution layer; a first semiconductor chip and a second semiconductor chip each on the first connection structure and each comprising connection pads electrically connected to the first connection via; an interconnection bridge below the first connection structure, the interconnection bridge comprising a connection electrode electrically connected to the first redistribution layer; an insulating material between the interconnection bridge and the first connection structure; and a second connection structure comprising a second redistribution layer below the interconnection bridge, and a second connection via connecting the first redistribution layer to the second redistribution layer, wherein the first connection via is tapered in a direction toward the connection pads of the first and the second semiconductor chips.

17. The semiconductor package of claim 16, wherein the first and the second semiconductor chips are electrically connected to each other through the interconnection bridge.

18. The semiconductor package of claim 16, wherein the connection electrode of the interconnection bridge is spaced apart from the first redistribution layer, and wherein the semiconductor package further comprises a connection member disposed between the connection electrode and the first redistribution layer.

19. The semiconductor package of claim 18, wherein the connection member is surrounded by the insulating material.

20. The semiconductor package of claim 16, further comprising an electrical connection metal below the second redistribution layer, wherein the electrical connection metal is electrically connected to the first and the second semiconductor chips through the second connection via.
